

Shihabul Ehsan

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EDUCATION

B.Sc. in Electrical and Electronic Engineering

Jan 2020 – Sep 2025

Khulna University of Engineering & Technology (KUET), Khulna, Bangladesh

CGPA: 3.25 / 4.00

Thesis: Design and Development of a Low-cost Dual-core Flight Controller for Quadcopter and Testing its Performance under Partial Propeller Disturbance Condition.

Achievements: Helped more than 100 students learn the basics of hardware and robotics by re-establishing the Robotics Club of KUET.

RESEARCH INTERESTS

Embedded Systems, Control Systems, Sensor Fusion, Robotics, UAV, ROS 2, Machine Learning, Deep Learning

PCB DESIGN EXPERIENCE

Remote Freelance PCB Layout Engineer

Dec 2025 – Present

Balboa Inspection & Maintenance, LLC · Contract (hourly)

Altium Designer

- Laid out a **1 Gbps 7-port Ethernet switch** with 5 integrated PHYs and 2 external PHYs over RGMII, using length-matched high-speed routing; manufactured and deployed in Balboa's Copperhead inspection robot.
- Laid out a **3-board Altium multi-board system** stacked inside a cylindrical enclosure, with dense component placement:
 - 300 V DC to 24 V DC isolated converter**
 - Power converter:** 24 V to 5 V, then 3.3, 2.5, 1.8, 1.2 and 1.1 V rails
 - Optical-to-Ethernet converter:** 2 optical + 2 copper ports, Ethernet switch IC and 2 PHYs
- Worked from client-supplied schematics in Altium Designer.

PROJECTS

Vector Drive V1: 3-Board BLDC Motor Drive System ([images](#))

- Designed a 2-wheel BLDC drive system for mini toys and robots.

Ultra-Low-Cost Cattle Health Monitoring System with LoRa Communication (client project) ([images](#))

- 450 μ A average current (calculated); runs about 6 months on one charge of a 1S 2200 mAh Li-ion battery.
- PCB manufactured by PCBWay.

ESP32 Mini Drone with PCB Frame ([images](#))

- Designed in KiCad; PCB manufacturing sponsored by JLCPCB ([flight video](#)).
- An earlier EasyEDA prototype with an MPU6050, milled in the KUET fabrication lab ([link](#)), also flew ([video](#)).

PCB Studio: Blender Extension for Realistic PCB Renders ([demo](#))

- Blender extension that generates realistic, product-level PCB renders with professional studio lighting in a few clicks; anyone can use it without learning Blender in detail ([GitHub](#)).

Smartphone-Controlled Mini Plane: ESP8266 Flight Controller PCBs ([flight video](#))

- Designed two ESP8266 flight controller PCBs in KiCad:
 - No-IMU version** ([images](#)): two unidirectional DC motor drivers, flown over Wi-Fi from a phone; the self-stable airframe glides without stabilization.
 - BMI088 IMU version** ([images](#)): four motor outputs; PCB sponsored by JLCPCB.

Advanced Line Follower Robot ([link](#))

- Built 10+ line follower robots and competed in 8 national competitions.
- Still working on bringing this product to market for kids ([link](#)).

Pipeline Corrosion Monitoring Robot ([link](#))

- Designed and developed the circuit and code of the vehicle.

Humanoid Eye Toy that can Interact with Kids ([link](#))

- This is my third-year project.

Flight Controller for Quadcopter from Scratch ([link](#))

- This is my thesis project.

Radio Controlled Glider Plane ([link](#))

- Implemented my quadcopter flight controller on this foam plane.

Development of an Extended Kalman Filter for Low-Cost Quadcopter

- Developed the extended Kalman filter in Python.
- Also developed an unscented Kalman filter for 6-DOF and 9-DOF IMUs.

SKILLS

Programming Languages: C, C++, Python, MATLAB

PCB Design: Altium Designer, KiCad **ADVANCED** · EasyEDA **FAMILIAR**

3D Design: Fusion 360

Prototyping: CNC PCB milling, 3D printing (own 3D printer)

Microcontrollers: Arduino, ESP32, STM32, Raspberry Pi

ML/DL Frameworks: TensorFlow, PyTorch, OpenCV

Robotics: ROS 2

LANGUAGES

English: Fluent

Bengali: Fluent

REFERENCES

Dr. Md. Shahjahan

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Khulna University of Engineering & Technology

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Relationship: Thesis Supervisor

Dr. Md. Jahirul Islam

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